

Alejandro Zumbado Vargas

Game & VR Developer

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SUMMARY

Computer Engineer (TEC, 2019) passionate about game development, from competitive programming and game design competitions to shipped products. Experience with Unity/C# (2D and 3D), multiplayer networking (Netcode for GameObjects + Relay), XR for Meta Quest and XREAL, procedural level generation, and even Game Boy homebrew development. Currently at Opal Interactive building XR simulators and kiosk/AR apps for clients; in parallel I publish my own games and I'm packaging my own networking layer as a standalone product.

SKILLS

Engine / Languages

Unity · C# · Shader Graph / ShaderLab · Python (game AI)

XR / Platforms

Meta Quest (VR) · XREAL (AR) · ARFoundation · A-Frame.js (web VR) · Android · WebGL · PC

Multiplayer

Netcode for GameObjects · Unity Relay · Photon · client-server architecture

AI / Tooling

Self-play game AI · procedural level generation · Unity editor tooling

WORK EXPERIENCE

Full-Stack / Game & VR Developer — Opal Interactive

Apr 2026 — Present

Studio that takes on projects from various companies and distributes them across its dev team

- HealthSim: Unity XR training simulator (XREAL glasses) for medical laser equipment, with remote expert–trainee sessions via Netcode for GameObjects + Unity Relay.
- totempanadol: Unity kiosk/AR app with thermal camera, face detection (ARFoundation), and QR code generation.
- Additional VR/Unity projects for studio clients: a VR minigolf game and a BrissaNova brand activation.

Full-Stack Developer, VR-focused (Jr/Mid) — WOW Emotions

Jan 2022 — Feb 2023

- Led end-to-end development of a VR-enabled website for the 'Doit' reconstruction project.
- Designed and built 'Giz': an immersive nature experience with 5+ VR levels.

Full-Stack Developer, 3D/Web VR-focused (Jr/Mid) — IMMERSO

Apr 2021 — Jan 2023

- Built interactive web and VR experiences with A-Frame.js: editors, maps, and database-driven content.

PERSONAL GAME & VR PROJECTS

Tinted Showdown — personal game — published 2025 — Present

2–4 player FFA multiplayer · Unity 6000 · PC / Android / WebGL

- Real-time color-matching combat game, playable at https://alejandrozumbado.github.io/tinted-showdown-build/.
- Networking built with Netcode for GameObjects + Unity Relay; fixed 8+ real production issues (WebGL endpoint selection, decompression fallback, Multiplayer Play Mode safety, RPC serialization limits).

QuickLobby — Netcode Relay Kit — personal product — in development Jul 2026 — Present

Drop-in join-code lobby system, preparing for release on Fab.com

- Extracted the networking layer from Tinted Showdown and packaged it as a standalone UPM asset with its own docs, changelog, and license.
- Fixes a WebGL-specific production bug (dynamic Relay endpoint selection) that silently breaks typical from-scratch implementations.

Cannons — personal game — published Aug 2025 — Present

Tower-defense · Unity · 500 levels · PC / Android

- Tower-defense game with progression across up to 500 levels and custom editor tooling for level generation, playable at https://alejandrozumbado.github.io/cannons-build/.
- Cross-platform support (PC mouse, Android touch) from a single codebase.

CannonsLevelGen — personal AI product Aug 2026 — Present

Self-play game AI in Python

- Python system that learns to play Cannons through self-play and uses what it learns to design new levels, running on GitHub Actions.

One-Way Royale — personal game — published 2021 — Present

Unity · Photon · Heroku backend

- Battle royale game with a self-built Heroku backend and Photon multiplayer.

TibetanCalligraphyVRapp — team project — VR 2025 — Present

Meta Quest · Unity · co-developed with a teammate

- VR Tibetan calligraphy app for Meta Quest, with brush-stroke interaction in an immersive environment.

ready-gamer (Game Boy homebrew) — personal project 2025 — Present

Homebrew C development for Game Boy (ZGB toolchain)

- Built a game for real Game Boy hardware in C, working within the original hardware's memory and processing constraints.

EDUCATION

Computer Engineering 2019

Tecnológico de Costa Rica (TEC)

High School Diploma 2015

Liceo San Rafael de Alajuela

LANGUAGES

Spanish — Native · English — Intermediate